# Credit Risk Analytics & NPA Prediction - Project Report

## Project Objective

The goal of this project is to analyze loan data, identify factors associated with credit risk, clean and prepare the dataset, and build insights related to Non-Performing Assets (NPA) and loan defaults.

## Key Tasks Performed

1. Data loading and memory optimization.
2. Data integration using loan master and related datasets.
3. Data quality checks and dirty data identification.
4. Missing value analysis and treatment.
5. Skewness analysis of numerical variables.
6. Exploratory Data Analysis (EDA).
7. Loan status and default analysis.
8. Credit score (CIBIL) based risk assessment.
9. NPA / default prediction analysis.

## Data Preparation

The notebook includes memory optimization, parquet conversion, data joining, orphan record checks, missing value handling, and data cleaning procedures to improve data quality.

## Exploratory Analysis

EDA was performed to understand loan behavior, default distribution, missing values, skewed variables, and relationships between borrower characteristics and loan performance.

## Key Insights

- Loan status was analyzed to identify performing and default accounts.
- Credit score (CIBIL) was evaluated as an important risk indicator.
- Missing values and data quality issues were identified and treated.
- Borrower and loan characteristics were studied to understand default patterns.

## Conclusion

The project demonstrates a complete credit risk analytics workflow including data preparation, EDA, risk assessment, and NPA prediction. Such analysis helps financial institutions identify high-risk borrowers and make better lending decisions.